# Statistics for Economics and Management

## Practical class #3 – Confidence Intervals for the Mean

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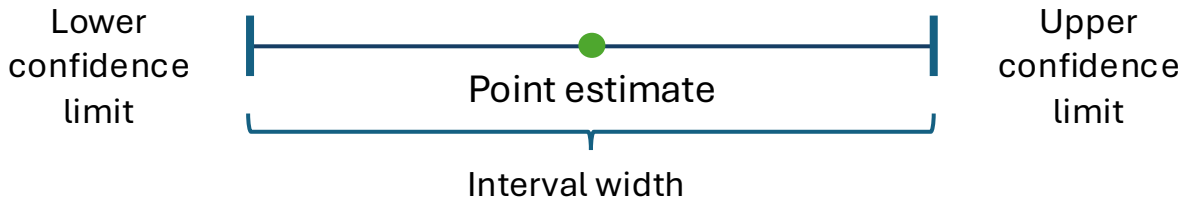
# Confidence Intervals

A confidence interval **estimator** for a population parameter is a rule for determining (based on sample information) an interval that is likely to include the parameter. It is based on a specific value obtained from the sample, $\hat{\theta}$, and takes into account the variation in our data through the margin of error.

Let $\theta$ be an unknown parameter. The $(1-\alpha)$ % confidence interval for $\theta$ defines two numbers between which the true population parameter is likely to fall. It is constructed as:

$$\boldsymbol{CI_{100(1-\alpha)\,\%,\theta} = \widehat{\theta} \pm \text{margin of error}}$$



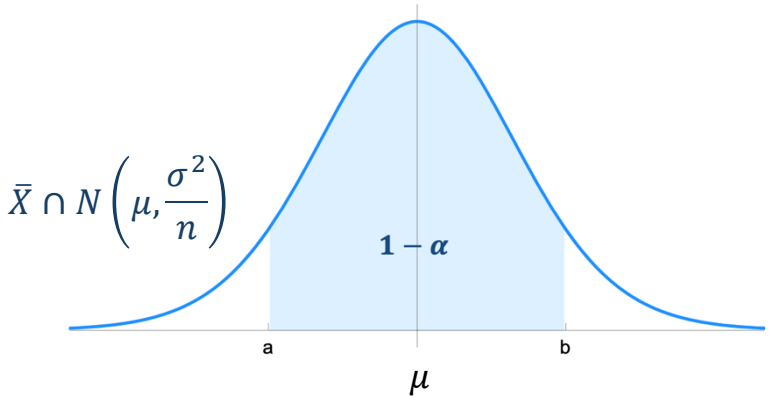

Width of the interval = 2 $\times$ margin of error

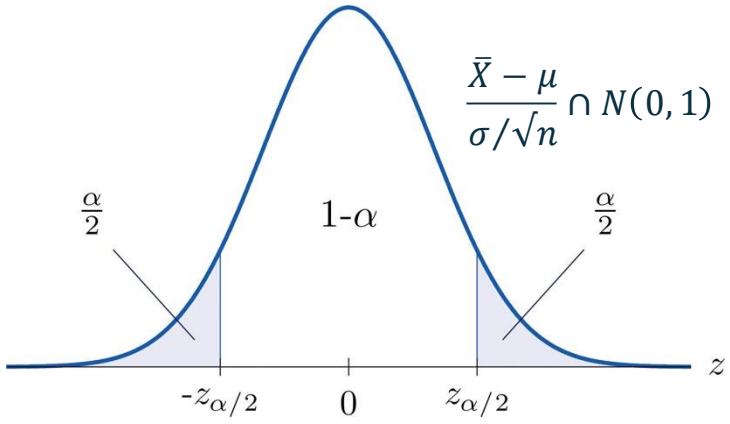
Note that $(1-\alpha)$ denotes **confidence level**, and $\alpha$ is the **significance level**.

# Confidence Interval for $\mu$ - $\sigma^2$ is known

- If our population follows a Normal distribution, we compute the $(1-\alpha)\%$ confidence interval for $\mu$, the population mean, as follows: $$\boldsymbol{CI_{(1-\alpha)\%,\mu} = \bar{X} \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}}$$

Notice that: $P\left(\bar{X} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}\right) = (1-\alpha)\% \quad \Leftrightarrow \quad P\left(-z_{\alpha/2} \leq \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{\alpha/2}\right) = (1-\alpha)\%$




- If our population does not follow a Normal distribution, then the Central Limit Theorem ensures that this formula is a (good) approximation, as long as our sample is large enough ($n > 30$).

# Confidence Interval for $\mu$ - $\sigma^2$ is unknown

Since $\boldsymbol{\sigma^2}$ **is unknown**, we estimate it with the sample variance, $\boldsymbol{s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$.

Notice that this introduces additional uncertainty into our formula, since the value of $\boldsymbol{s^2}$ will vary from sample to sample.

- If our population follows a Normal distribution, we compute the $(1-\alpha)\%$ confidence interval for $\mu$ as follows:

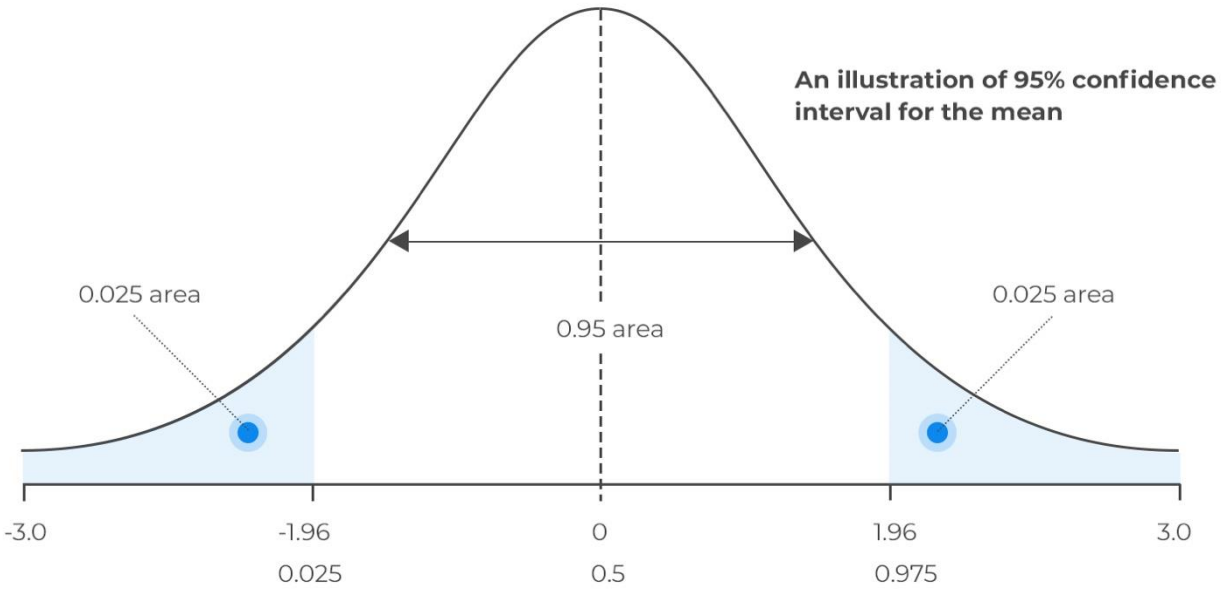
$$\boldsymbol{CI_{(1-\alpha)\%,\mu} = \bar{X} \pm t_{n-1,\alpha/2} \times \frac{s}{\sqrt{n}}}$$

- If our population does not follow a Normal distribution, then the Central Limit Theorem ensures that this formula is a (good) approximation, as long as our sample is large enough ($n > 30$).

Notice that here we use the **Student's t distribution**, with $\mathbf{n-1}$ **degrees of freedom**, to obtain our reliability factor. It can be empirically shown that the Student's t distribution works best here.

# Confidence Intervals

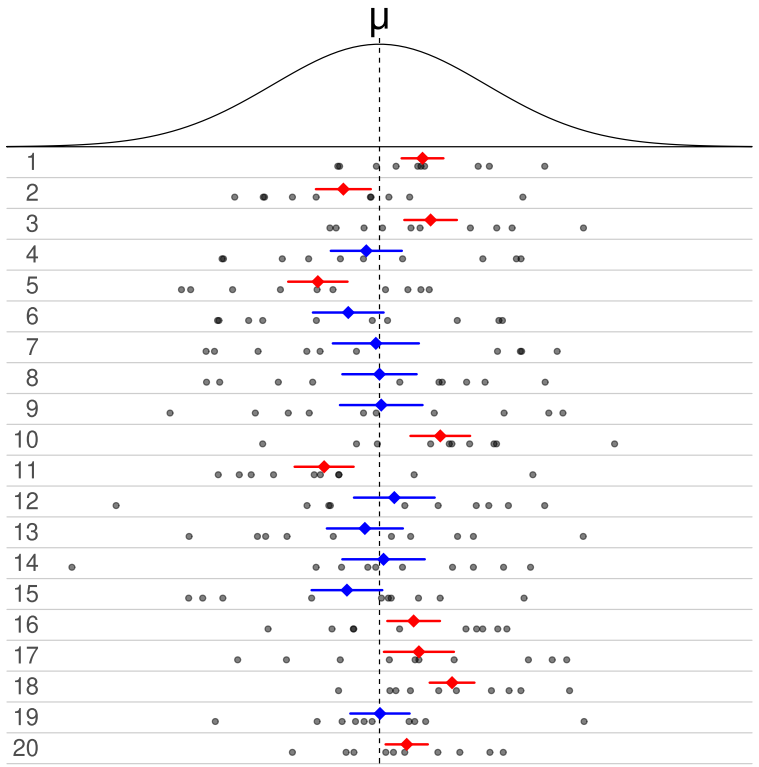
Note that there is a **trade-off between $\alpha$ and the width of the interval**. A higher confidence level (or equivalently, a lower significance level, $\alpha$) will produce wider confidence intervals. Therefore, the more confident we want to be that our interval contains the true population parameter, the less precise will the information contained in such interval be.



# Interpretation

Two **correct** interpretations for a confidence interval for $\mu$:

- If we repeatedly take random samples from our population, $(1-\alpha)\%$ of the confidence intervals obtained using our formula will contain the unknown population mean $(\mu)$ .
- For a (theoretical) **random** sample, there is a $(1-\alpha)\%$ probability that the confidence interval using the formula will contain the unknown population mean $\mu$.




For **one specific confidence interval**, which we compute based on one specific sample, our population parameter will either be contained or not contained in the interval. There is **no probability** associated with the parameter being within the interval. The best we can say is that we are $(1-\alpha)\%$ **confident** that the true parameter lies within the confidence interval we obtained.

# Workbook, exercise 4.6

A process for producing bricks is known to give output whose weights are normally distributed with standard deviation 0.12 Kg. A random sample ($n = 16$) from today's output had a mean weight of 4.07 Kg.

(a) Find a 99% confidence interval for the mean weight of all bricks produced this day.

(b) Without doing the calculations, explain whether a 95% confidence interval for the population mean would be wider than, narrower than, or the same width as that found in part (a).

(c) It is decided that tomorrow a sample of 20 bricks will be taken. Without doing the calculations, explain whether a correctly calculated 99% confidence interval for the mean weight of tomorrow's output would be wider than, narrower than, or the same width as that found in part (a).

(d) Suppose that the population standard deviation for today's output is 0.15 Kg (not 0.12 Kg). Without doing the calculations, explain whether a correctly calculated 99% confidence interval for the mean weight of today's output would be wider, narrower, or the same width as that found in part (a).

# Workbook, exercise 4.11

A consumer association decided to investigate whether the soda cans sold by various brands are correctly filled. Let $X$ equal the amount of refrigerant in centiliters actually existing in a can of New-Cola brand chosen at random. Suppose $X$ has a normal distribution with mean $\mu$ and standard deviation $\sigma$. With the aim of estimating the value of $\mu$, a random sample of $n = 9$ cans will be selected. Answer the following questions by presenting and justifying all the necessary calculations.

(a) Suppose the sample mean was $\bar{x} = 38$. Knowing that $\sigma$ = 2, propose a 95% confidence interval for $\mu$.

(b) If the value of $\sigma$ was unknown and had to be estimated, how would the amplitude of the new confidence interval change? Explain.

(c) One of your colleagues proposes the following confidence interval for $\mu$: $(\bar{X} - 1.1, \bar{X} + 1.1)$, where $\bar{X}$ denotes the sample mean. Suppose $\sigma$ is known and equals 2.

i. What is the confidence level of the proposed confidence interval?

ii. To obtain a 95% confidence level using the proposed confidence interval, what should be the value of $n$?

# Workbook, exercise 4.5

We selected a random sample of 25 students from the population. Below, we have some descriptive statistics regarding the number of hours each student slept the night before:

Minimum = 5.8 Maximum = 9.5 Average = 7.0 Median = 6.5 Standard Deviation = 1.0

Answer the following questions assuming that the population distribution is normal.

a) Suppose we knew that the standard deviation of hours slept by all students in the population was 0.5. What is the 95% confidence interval for the average hours slept by all students in the population? If you want the margin of error of the 95% confidence interval to be 0.1 hours, what should be the sample size?

b) Suppose now that the variance of hours slept by all students in the population was unknown. Using the statistics presented above regarding the sample of 25 students, what is the 95% confidence interval for the average hours slept by all students in the population?

# Workbook, exercise 4.5 - continued

We selected a random sample of 25 students from the population. Below, we have some descriptive statistics regarding the number of hours each student slept the night before:

Minimum = 5.8 Maximum = 9.5 Average = 7.0 Median = 6.5 Standard Deviation = 1.0

Answer the following questions assuming that the population distribution is normal.

c) What is the meaning of a 95% confidence interval for the average hours slept by all students in the population?

   i. 95% of the students in the population slept a number of hours inside that interval
   ii. 95% of the students in the sample slept a number of hours inside that interval
   iii. the probability that a random sample results in a sample mean with a value inside that interval is equal to 95%
   iv. all previous statements are incorrect